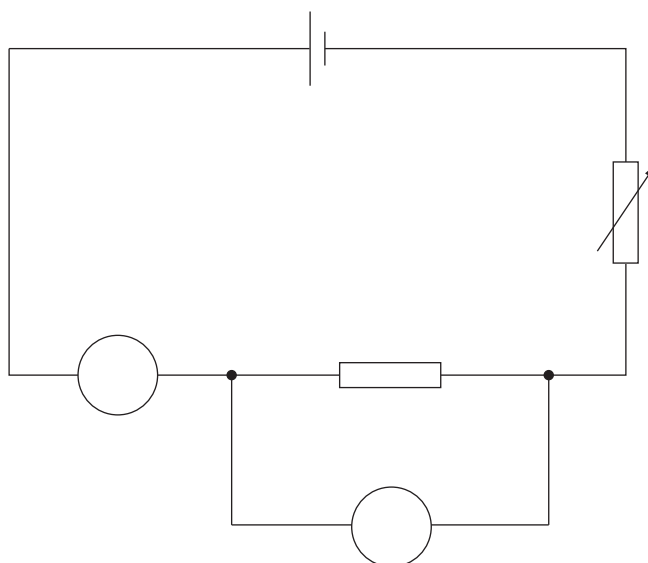


3. Students investigate the I - V characteristic of a resistor. They set up the following circuit.



- (a) **Complete the circuit** by labelling the ammeter and voltmeter with their correct symbols. [1]

- (b) The students' results are given in the table below.

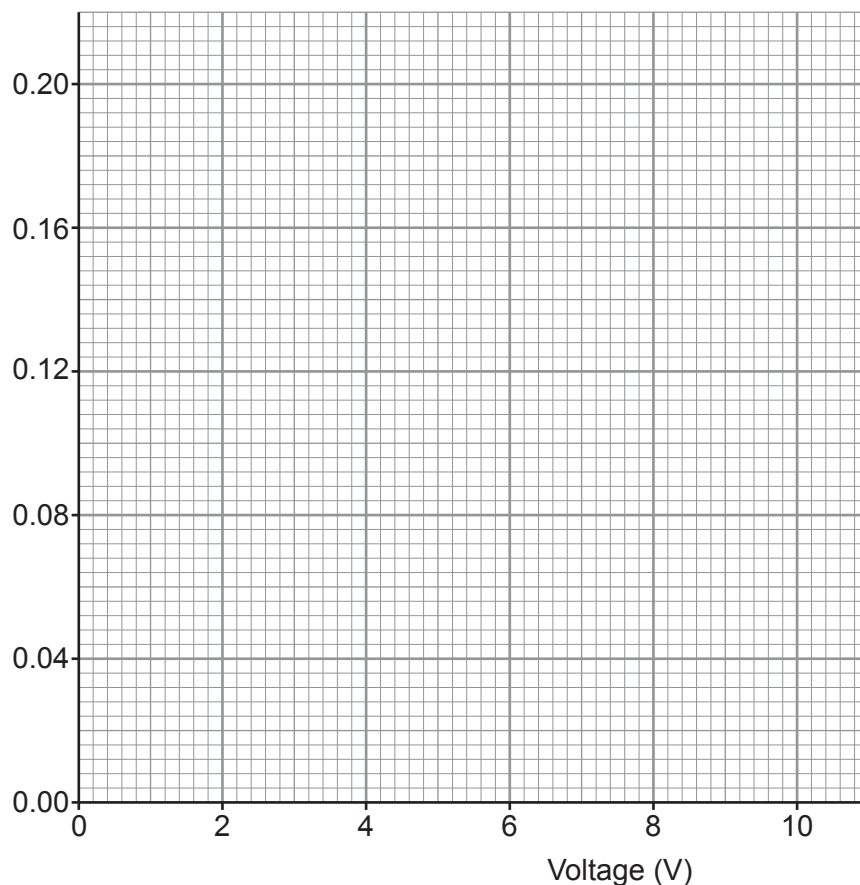
Voltage (V)	Current (A)
0	0.00
2	0.04
3	0.06
5	0.10
8	0.16
10	0.20



- (i) Plot the data on the grid below and draw a suitable line.

[3]

Current (A)



- (ii) Use the graph to determine the current at 4 V.

[1]

current = A

- (iii) Use the equation:

$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to determine the resistance of the resistor at 4 V.

[2]

resistance = Ω



Examiner
only

- (iv) Sally suggests that the resistance of the resistor is constant.
Use the graph to explain whether you agree. [1]

.....

.....

- (v) In the circuit, the resistance of the variable resistor is set at 25Ω .
Use your answer to part (b)(iii) and an equation from page 2 to calculate the total
resistance of the circuit. [2]

total resistance = Ω

10

